

Report

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Introduction

This report investigates patterns in haunted place reports across the United States by integrating multimodal datasets and visualization techniques. Building on insights from previous assignments, our team used D3 visualizations, Apache Solr queries, mapping, and image analysis tools to examine how geographic, cultural, psychological, and environmental factors shape paranormal narratives. Each method added context: bubble maps linked hauntings to alcohol-related deaths; radial charts and Solr explored religious ties; scatter plots examined daylight and violence; and circle packing maps showed apparition diversity in cities. We also analyzed AI-generated image analysis revealed symbolic visual patterns, while mapping workflows revealed spatial clustering and location-specific themes. This report presents our key findings, including geographic clustering, religious and symbolic associations, and the influence of emotional or environmental conditions.

D3 Visualizations

Bubble Map

We chose the D3 bubble map to visually highlight the overlapping patterns between haunted place density and alcohol-related deaths across the US, using key features from assignment 1. Each bubble's size represents the number of reported hauntings (`haunted_count`), while its color intensity maps to the average alcohol related death rate (`avg_total_deaths`). Darker red shades indicate higher alcohol-related fatalities, while larger bubbles indicate more frequent hauntings. From the bubble map, we can observe that states like California, Texas, and Florida show both large, dark red bubbles, thus signaling high paranormal activity and high alcohol mortality. This reinforces our insight that regions with greater haunted activity are correlated to alcohol-related deaths. In contrast, some parts of the Northeast and Midwest display large pale-shaded bubbles, suggesting high haunting counts occur even in areas with moderate alcohol-related deaths. This could potentially mean there were other cultural or historical influences. Overall, the bubble map clearly ties together both haunted and alcohol-related features and generally supports the hypothesis made in assignment 1.

Radial Bar Chart

We chose a radial stacked bar chart because it builds directly on the findings from A1 and A2, offering a clearer comparison of haunted place proximity to worship sites and apparition type diversity across religions. While A1 showed that hauntings cluster near religious centers, especially Christian and Jewish sites, the radial chart adds more specificity by quantifying median distances and breaking down apparition types by religion. Christianity appears spatially mid-ranged in distance compared to other religions and shows the greatest diversity in apparition types, likely because it is the most widespread in the dataset. The chart also confirms the A1 insight that Ghosts and Spirits are the most universally reported apparitions, appearing across all traditions, which may be tied to the inter-religious beliefs in "souls," "spiritual energy," and the afterlife. In contrast, Demons and Entity/Light apparitions are reported solely near Christian sites, which makes sense because these types of supernatural beings are heavily rooted in Christian theology and popularized through Western media. Multiple Apparitions appear more near religions with pluralistic or syncretic views (e.g., Jewish, Muslim, Scientologist). Finally, religions with only "Unknown" apparitions likely reflect small sample sizes or cultural framing differences.

Scatterplot:

We chose a D3 scatterplot to visualize the relationship between daylight duration and apparition types in reported hauntings, building on earlier analysis of environmental and psychological factors influencing haunted place patterns. Each dot in the scatter plot represents a reported haunting, positioned

horizontally by the number of daylight hours and vertically by the primary apparition type. The color of each dot indicates whether the event was classified as violent (red) or non-violent (blue), based on event descriptors such as “attack,” “possession,” or “murder.” The scatter plot reveals that Spirits and Orbs are more commonly reported during longer daylight periods, suggesting that lighter conditions may influence the types of apparitions perceived or recorded. In contrast, Ghosts and Demons show a broader spread across all daylight durations, indicating that sightings of these apparition types are less influenced by lighting and may be tied to psychological expectations or cultural narratives that persist regardless of environmental conditions. Violent hauntings appear distributed evenly across the daylight spectrum, implying that the severity of the encounter is not strongly linked to time of day. This visualization deepens the understanding from earlier analyses by quantifying and visualizing how daylight and apparition type interact, suggesting that while the environment plays a role, other beliefs about specific supernatural entities may be a stronger driver in shaping haunted narratives.

Haunted Flight Explorer

We created an interactive scatter map to visualize the airport and air traffic geospatial data we added in assignments one and two. This map plots the locations, nearby airports, and intersecting flight routes for each haunted place. Users can filter, color code, and scale the size of haunted places by the features we added in assignment one. The goal of this visualization is to further contextualize the haunted places dataset and allow users to dive deeper into the stories they find interesting. For example, In the 1960s, a family of three died in a private plane crash at “Airplane Hollow” located in Nelsonville, Ohio. We observed no nearby airports, and that the crash overlaps with flight routes from Pittsburgh to Memphis. Perhaps, our unfortunate victims ran out of fuel on their way to Graceland before they met their untimely death. The skeptics of the world can also use this visualization to ruin other people’s fun. For example, the “Marfa Lights”, located between Marfa and Alpine Texas, is a UFO tourist attraction where spectators spot mysterious floating lights in the night sky. Coincidentally, this location overlaps perfectly with flights from Laredo, TX to Las Vegas, NV. Perhaps these “unidentified flying objects” are really planes full of “unidentified flying opportunists”. All jokes aside, playing with the tool is more fun than writing a report on it, so we encourage you to try out the [visualization](#) for yourself.

Circle Packing Map

We chose the D3 circle packing map to show how different apparition types manifest across U.S. cities, building on the classification work from Assignment 1 and the geographic analysis from Assignment 2. Large circles represent apparition types—like Ghost, Spirit, or Unknown—while smaller nested circles indicate cities reporting sightings. City size reflects report volume, with labels added for cities with more than 10 sightings. Solid circles indicate cities tied to a single type; bold outlines show cities reporting multiple types. The map clearly shows that Ghosts and Spirits dominate the dataset, appearing frequently and across a wide geographic range. This supports our Assignment 1 findings, where metadata analysis also revealed these as the top apparition categories. Cities like Los Angeles, San Antonio, and Columbus stand out not only for their high number of sightings but also for reporting a broader mix of apparition types. This aligns with Assignment 2 findings that hauntings cluster in urban, culturally rich areas with dense populations and more storytelling traditions. Overall, the map helps visualize our previous findings, showing that apparition frequency and diversity are shaped by both entity classification and urban cultural context.

Apache SOLR Observations

Using Apache Solr queries, we explored the relationship between hauntings and alcohol-related deaths. By filtering for locations with both high haunted activity (`haunted_count ≥ 3`) and high average alcohol-related deaths (`avg_total_deaths ≥ 8000`), we identified cities like Antioch (CA), Austin (TX), and Missoula (MT) as key overlap zones. We also examined the time of day of hauntings in these high-death areas. The data showed a distinct contrast, where 607 hauntings occurred in the Evening, compared to

only 56 in the Morning and 7 in the Afternoon. This supports the idea that alcohol consumption, which tends to increase at night, may impair perception and contribute to more supernatural reports during evening hours.

In addition to behavioral and environmental factors like alcohol use, we also explored how religious contexts shape the way hauntings are reported and interpreted. While A1 clustering showed hauntings cluster near worship sites and A3's radial chart revealed variation in apparition diversity across religions, Solr queries added specificity by examining apparition diversity, mental health, and location type. Christian-affiliated sites continued to dominate both spatially and culturally. Our query for Religion_Intersection:christian (see Appendix, Screenshot 1) confirmed that 87.4% (874/1000) of entries in the stratified sample were Christian, aligning with our earlier clustering and visualization findings. A follow-up query (Religion_Intersection:christian AND Apparition_Type:*, see Appendix, Screenshot 2) confirmed Christianity's broad range of apparitions (Ghosts, Spirits, Demons, Orbs, Entity/Light), validating A3's diversity patterns. Solr also revealed that Christian-linked hauntings often occurred in emotionally charged, culturally significant locations like historic homes, inns, and theaters (Appendix, Screenshot 1), expanding on our earlier observations about symbolic and historical framing. Additional queries (e.g., Religion_Intersection:christian AND Depression_Prevalence:[0.2 TO *], Appendix, Screenshot 3) highlighted that haunted places near Christian worship sites with high depression rates tended to report intense or ambiguous phenomena: disembodied voices, flickering lights, electrical malfunctions, moving objects, and trauma-linked narratives (Appendix, Screenshot 2). These results build on A2's insight that emotional distress may heighten the vagueness, intensity, and symbolic nature of haunting accounts, especially in religious contexts.

Using Solr-based queries, we specifically investigated major urban areas to better understand how haunting activity manifests in densely populated settings. In Los Angeles, we identified 80 haunted places, many of which were associated with prominent historical locations such as the Ambassador Hotel. In New York, we located 125 haunted places, reflecting a broad diversity of site types, from historic buildings to culturally significant neighborhoods, further reinforcing the strong relationship between dense urban environments and high reported haunting activity. Expanding the analysis to other major cities, Chicago revealed 30 haunted places, with many entries linked to iconic sites like Archer Woods Cemetery, where reports frequently documented both audio and visual supernatural evidence. Similarly, in Baltimore, 30 haunted places were found, often located in transitional spaces such as former trailer parks and industrial zones, suggesting the influence of community disruption and urban change on the formation of haunting narratives. In Atlanta, Solr queries returned 11 haunted places, commonly centered around rural-adjacent locations and historic structures with traumatic associations, such as disease outbreaks and early settlement deaths. Reports from these sites frequently included both visual and auditory phenomena. Finally, in Birmingham, we identified 8 haunted places, largely tied to early 20th-century residential homes.

Visual evidence was consistently reported in these cases, while audio phenomena were much less common. These findings align with the broader pattern observed earlier: larger urban areas like New York and Los Angeles yield higher reported haunting counts, consistent with expectations based on historical layering, building density, and collective cultural memory. Conversely, smaller but historically significant urban areas like Atlanta and Birmingham highlight how localized narratives can concentrate haunting reports around a few culturally resonant sites.

In addition to city-level analysis, we examined how different types of supernatural evidence were distributed across the full haunted places dataset. A query for Visual_Evidence:true returned 7,208 results, indicating that approximately 65% of all reported hauntings involved some form of visual phenomena, such as sightings of apparitions, shadows, or unexplained lights. A corresponding query for Audio_Evidence:true returned 5,413 results, suggesting that approximately 49% of hauntings included auditory phenomena such as disembodied voices, footsteps, or mysterious noises.

The higher prevalence of visual evidence over audio evidence supports psychological theories that emphasize the primacy of vision in human perception, particularly in ambiguous, emotionally charged, and low-light environments where the imagination can easily heighten or distort sensory

experiences. This observation aligns with earlier findings regarding the dominance of visually reported supernatural events, particularly in contexts where atmospheric or psychological conditions may amplify suggestibility.

Image Space

We successfully ingested a subset of 1000 images from the Haunted Places dataset into ImageSpace using the Docker-based deployment approach outlined in the ImageCat Quick Start Guide. The process involved setting up Docker containers for the ImageSpace ecosystem, including Girder, MongoDB, and Solr, and mounting our local image directory for in-place access. Apache Tika was used to extract rich metadata including EXIF data and image properties as part of our ingestion pipeline. Once the images were indexed in Solr with a schema customized for haunted places attributes, the SMQTK components were configured to generate feature vectors using pre-trained CNN models and build a Locality-Sensitive Hashing (LSH) index for efficient similarity searching. This configuration allowed for browsing and searching the haunted places images through the ImageSpace interface, where similar images could be found using both content-based image retrieval (through SMQTK's visual similarity capabilities) and forensic analysis of image attributes. The similarity searches revealed several significant patterns in the dataset: architectural similarities grouped Victorian mansions, gothic churches, and abandoned asylums together; environmental factors like fog, mist, and lighting conditions created natural clusters; and thematic elements related to supernatural features showed correlation in the similarity space. Some of these elements were hard to describe even for our team, and when we saw the images side by side, we understood even more.

In parallel, members of our team developed a complementary pipeline using Python scripts, OpenCV-based k-means clustering, and an Elasticsearch backend to overcome setup constraints posed by legacy ImageSpace dependencies. We clustered the 1000 images into three groups ($k=3$), chosen to balance interpretability with meaningful visual differentiation across the dataset. These clusters revealed structural themes and lighting consistencies not captured by raw feature vectors alone. The assigned clusters were visualized using bar plots, and the top-5 nearest neighbors per query image were displayed to validate the internal coherence of each group. Image metadata and cluster labels were indexed in Elasticsearch, replicating similarity retrieval functionality without direct use of ImageCat outputs. This dual-track strategy highlighted both the challenges and the adaptability required in visual search projects, demonstrating the value of hybrid methods for image understanding.

Geoparser Findings

The original GeoParser tool included in the assignment was outdated and no longer functional. It relied on deprecated Django and Solr components that were incompatible with current Python environments. Despite efforts to configure the application, it failed to run properly, and the web interface could not handle file uploads or parse the dataset. As a result, we developed a custom GeoParser pipeline to extract and map geographic data from our haunted places dataset. This custom pipeline began by using spaCy's Named Entity Recognition (NER) model to identify location entities (GPEs) in over 10,000 haunted descriptions. Once extracted, the location names were passed to the geopy library, which geocoded them using Nominatim. To handle limitations of the free API, the script included retry and timeout logic and enforced rate limiting with a one-second delay between requests. The output included more than 2,000 successfully geocoded haunted places, which were saved to a structured CSV (parsed_haunted_locations.csv) containing original text, extracted location names, and their corresponding latitude and longitude values. Finally, we used the folium library to create an interactive map of the geocoded points. Each marker on the map displays the original haunted description and its location, enabling users to explore geographic patterns in the haunting narratives. This approach not only replicated the intended functionality of the original GeoParser, but scaled it up to handle the full dataset while producing outputs that were more structured, visual, and useful for analysis.'

Using our Geoparser, we extracted over 2,000 unique, geocodable location references from the haunted descriptions dataset and visualized them using folium. The most common types of location data included city names, street names, and local landmarks such as cemeteries, hospitals, and schools. These ranged from precise urban institutions to more vague or rural descriptors like abandoned houses or bridges in the woods. While spaCy's Named Entity Recognition (NER) model effectively identified most geopolitical entities, some ambiguous place names—such as “Washington” or “Springfield”—required additional disambiguation, which occasionally resulted in noise during geocoding.

The interactive map revealed several notable spatial patterns that expanded upon earlier findings from Assignment 1. We observed strong geographic clustering of hauntings in high-population regions such as the Midwest, Northeast, and parts of California, reinforcing our previous conclusion that urban areas tend to have more reported supernatural activity. Additionally, descriptions that included richer metadata, such as compound place names or institutional references, were more likely to be successfully geocoded, suggesting a correlation between narrative specificity and extractability. The map also highlighted spatial trends not previously visible in the textual analysis alone, such as a concentration of hauntings near former transportation routes or institutional sites, pointing to potential historical or social factors shaping the distribution of ghost reports.

Conclusion

Our analyses confirm that haunted place narratives are shaped by spatial, cultural, psychological, and environmental factors. We found that hauntings often cluster near religious institutions, particularly Christian ones, and that the diversity of apparition types reflects the theological and symbolic nuances of local belief systems. Visualizations reinforced the dominance of Ghosts and Spirits and highlighted how alcohol use, urban density, and even daylight hours may influence how hauntings are perceived and reported. Through Apache Solr, we explored emotional tone, location type, and mental health correlations, finding that trauma, depression, and symbolic resonance amplify haunting narratives, especially in historic and religiously affiliated sites. Our GeoParser revealed geographic patterns and location-specific storytelling, while our image similarity work uncovered thematic consistency in haunted visuals. Together, these methods deepen our understanding of how hauntings are perceived, interpreted, and narrated across diverse cultural and spatial contexts in the U.S.

Appendix: Solr Query Screenshots.

This appendix contains supporting screenshots for the Solr queries referenced throughout the report. Each screenshot includes either the query performed or sample results illustrating key findings discussed in the main analysis.

Screenshot 1A: Query - Christianity Dominates Spatially

Query: *Religion_Intersection:christian*

Shows numFound = 874 (out of 1000) haunted places linked to Christian worship sites.

```
{
  "responseHeader": {
    "status": 0,
    "QTime": 4,
    "params": {
      "q": "Religion_Intersection:christian",
      "indent": "on",
      "wt": "json",
      "_": "1745005043581"
    }
  },
  "response": { "numFound": 874, "start": 0, "docs": [
```

Screenshot 1B–1F: Sample Locations for Christian-Affiliated Hauntings

Query: *Religion_Intersection:christian*

Examples of locations associated with Christian-linked hauntings:

```
"Location": ["Alexander Inn"],
"Location": ["Eaview twin theaters"],
```

```
"Location":["Old Pea ridge Mine"],  
"Location":["The Cliff House"],  
"Location":["The Hemingway House"],
```

Screenshot 2A: Query - Apparition Diversity at Christian Sites

Query: *Religion_Intersection:christian AND Apparition_Type:**

Shows results containing apparition types such as Orb, Ghost, Unknown.

```
"q":"Religion_Intersection:christian AND Apparition_Type:*",  
... ..
```

Screenshot 2B–2C: Sample Apparition Types - Orb and Ghost

Query: *Religion_Intersection:christian AND Apparition_Type:**

Examples of apparition results:

```
"Apparition_Type":["Orb"],  
"Apparition_Type":["Ghost"],
```

Screenshot 3A: Query - Christianity + Elevated Depression Prevalence

Query: *Religion_Intersection:christian AND Depression_Prevalence:[0.2 TO *]*

Identifies haunted places near Christian worship sites with above-average depression prevalence.

```
"q":"Religion_Intersection:christian AND Depression_Prevalence:[0.2
```

Screenshot 3B–3E: Sample Haunting Descriptions and Event Types

Query: *Religion_Intersection:christian AND Depression_Prevalence:[0.2 TO *]*

Sample Event_Types:

```
"Event_Type":["Supernatural"],  
"Event_Type":["Electronic_Malfunction | Supernatural"],  
... ..  
"Event_Type":["Violence"],  
... ..  
"Event_Type":["Supernatural | Violence"],  
... ..  
"Event_Type":["Accident/Disaster"],  
... ..
```

Screenshot 4A-B: Location

Query: *City:Los Angeles OR City:New York*

```

},
"response":{
  "numFound":8,
  "start":0,
  "numFoundExact":true,
  "docs":[{
    "Haunted_Places_Id":36,
    "City":["Birmingham"],
    "Country":["United States"],
    "response":{
      "numFound":80,
      "start":0,
      "numFoundExact":true,
      "docs":[{
        "Haunted_Places_Id":2796,
        "City":["Los Angeles"],
        "Country":["United States"],
        "Description":["Last year, during the fi
        "Location":["Ambassador Hotel"],
        "State":["California"],

```

Screenshot 5A-B: Audio-Visual Evidence

Query: *Audio_Evidence:True, Visual_Evidence:True*

q

Audio_Evidence:true

q.op

OR

fq

sort

start, rows

010

fl

df

paramset(s)

Select paramset(s)...

responseheader :

"status":0,

"QTime":21,

"params":{

"q":"Audio_Evidence:true",

"indent":"true",

"q.op":"OR",

"useParams":"","

"_":"1745871256424"

}

},

"response":{

"numFound":5413,

"start":0,

"numFoundExact":true,

"docs":[{

"Haunted_Places_Id":0,

"City":["Ada"],

"Country":["United States"],

"Description":["Ada witch – Sometime

"Location":["Ada Cemetery"],

"State":["Michigan"],

"State_Abbrev":["MI"],

q

Visual_Evidence:true

q.op

OR

fq

sort

start, rows

010

fl

df

paramset(s)

Select paramset(s)...

"status":0,

"QTime":24,

"params":{

"q":"Visual_Evidence:true",

"indent":"true",

"q.op":"OR",

"useParams":"","

"_":"1745871256424"

}

},

"response":{

"numFound":7208,

"start":0,

"numFoundExact":true,

"docs":[{

"Haunted_Places_Id":0,

"City":["Ada"],

"Country":["United States"],

"Description":["Ada witch – Someti

"Location":["Ada Cemetery"],

"State":["Michigan"],

"State_Abbrev":["MI"],

Screenshot 6: Time of Day and Deaths Evidence

Query: *most_common_time_of_day:Evening AND avg_total_deaths:[8000 TO *]*



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Request-Handler (qt) /select
common
q most_common_time_of_day:evening AND
q.op OR
fq
sort
start, rows 0 1000
fl most_common_time_of_day
df
paramset(s) Select paramset(s)...
wt
indent on
debugQuery
defType

http://localhost:8983/solr/hauntedplaces/select?fl=most_common_time_of_day&indent=
{
"responseHeader":{
"status":0,
"QTime":2,
"params":{
"q":"most_common_time_of_day:evening AND avg_total_deaths:[8000 TO *]"
"indent":"true",
"fl":"most_common_time_of_day",
"q.op":"OR",
"rows":"1000",
"useParams":"","
_: "1744334583642"
}
},
"response":{
"numFound":607,
"start":0,
"numFoundExact":true,
"docs":[{
"most_common_time_of_day":["Evening"]
},{
"most_common_time_of_day":["Evening"]
},{
"most_common_time_of_day":["Evening"]
},{
"most_common_time_of_day":["Evening"]
},{
"most_common_time_of_day":["Evening"]
},{
"most_common_time_of_day":["Evening"]
},{
"most_common_time_of_day":["Evening"]
},{
"most_common_time_of_day":["Evening"]
},{
"most_common_time_of_day":["Evening"]
},{
"most_common_time_of_day":["Evening"]
}
}
}

Screenshot 7: Haunted Sightings and Alcohol

Query: *haunted_count:[3 TO *] AND avg_total_deaths:[8000 TO *]*



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Request-Handler (qt) /select
common
q haunted_count:[3 TO *] AND avg_total_deaths:
q.op OR
fq
sort
start, rows 0 1000
fl
df
paramset(s) Select paramset(s)...
wt
indent on
debugQuery
defType
hl
facet
spatial
spellcheck
Raw Query Parameters
JSON Query

http://localhost:8983/solr/hauntedplaces/select?indent=true&q.op=OR&q=haunted_cou
{
"responseHeader":{
"status":0,
"QTime":0,
"params":{
"q":"haunted_count:[3 TO *] AND avg_total_deaths:[8000 TO *]",
"indent":"true",
"q.op":"OR",
"rows":"1000",
"useParams":"","
_: "1744334583642"
}
},
"response":{
"numFound":23,
"start":0,
"numFoundExact":true,
"docs":[{
"id":"doc_298",
"city":["Antioch"],
"state":["CA"],
"latitude":[37.9368191],
"longitude":[-121.7966315],
"haunted_count":[3],
"avg_total_deaths":[15443.0],
"avg_percent_under_21":[2.5],
"most_common_time_of_day":["Unknown"],
"_version_":1829067368898232323,
"_root_":"doc_298"
},{
"id":"doc_299",
"city":["Antioch"],
"state":["CA"],
"latitude":[37.971782],
"longitude":[-121.861398],
"haunted_count":[3],
"avg_total_deaths":[15443.0],
"avg_percent_under_21":[2.5],
"most_common_time_of_day":["Unknown"],
"_version_":1829067368898232324,
"_root_":"doc_299"
},{
"id":"doc_352",